Stellar WiFi Analyzer

User Guide

V1.1.5-build-9.10

Purpose-built for Alcatel-Lucent Enterprise Stellar WiFi Access Points

Enterprise Mode • Cluster Mode

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About the Author

Fernando Rivasplata

Certifications	Description
CWNA	Certified Wireless Network Administrator
CWAP	Certified Wireless Analysis Professional
CWSP	Certified Wireless Security Professional

✝️ Faith & Family

Fernando is a devoted follower of Jesus Christ, saved by the grace of God. He is a blessed husband and father — values that guide both his personal life and his professional integrity.

■ AI & Automation Enthusiast

Fernando is a passionate AI trainer enthusiast and a Network Automation and Security Specialist who continuously explores how artificial intelligence and automation can transform network operations and simplify complex troubleshooting workflows.

■ WiFi Expertise — 20+ Years

With more than 20 years of hands-on experience in Wi-Fi networking, Fernando brings a deep theoretical and practical understanding of IEEE 802.11 standards, RF behavior, security protocols, and real-world wireless deployments. His CWNA, CWAP, and CWSP certifications validate his expertise across all key dimensions of enterprise wireless.

■ Alcatel-Lucent Enterprise — 15 Years

Fernando serves as a Network Architect within Alcatel-Lucent Enterprise (ALE), where he has spent the last 15 years building deep expertise in Stellar WiFi — ALE’s enterprise wireless platform. His intimate knowledge of how Stellar WiFi access points operate internally — from firmware logs to RF behavior — is the foundation upon which Stellar WiFi Analyzer was built.

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1 Introduction — What Is Stellar WiFi Analyzer?

Stellar WiFi Analyzer is a comprehensive, browser-based network troubleshooting and forensic analysis tool purpose-built for Stellar WiFi Access Points, the Stellar Access Points are property of Alcatel-Lucent Enterprise (ALE). It connects directly to one or more APs via SSH and provides a rich set of real-time diagnostic tools — all from a single web browser tab, with no agents or additional software required on the APs themselves.

The tool was designed and developed by Fernando Rivasplata, a Network Architect at ALE with +20 years of deep expertise in WiFi. It encodes years of operational knowledge about how Stellar APs behave internally, making expert-level troubleshooting accessible to any engineer who uses it.

Key Capabilities

Function	What It Does
Device Signal Monitor	Live RSSI, noise, SNR, retransmissions and CU with CWAP-level RF analysis, Follow Roaming, and Anomaly detection
Channel Utilization Monitor	Real-time per-band airtime congestion monitoring across 2.4 / 5 / 6 GHz with continuous trending
CCI & ACI Detection	Co-channel and adjacent-channel interference scan with signal-weighted impact scoring and per-channel noise floor
Locate a Device	Find any connected client across all APs by MAC, IP, hostname, RSSI, or SNR
RTT & Jitter Monitor	Continuous round-trip latency and jitter measurement from AP to any destination host
Multicast/Broadcast Monitor	Real-time multicast and broadcast rate monitoring with dangerous-rate alerting and remediation guidance
Ethernet Capture	Run tcpdump directly on AP ethernet interfaces and download .pcap files
AP Information	Hardware, firmware, radio, SSID, and health snapshot for any AP
Config Files Browser	Browse, view, and download AP configuration and system files directly from the SSH shell
Roaming Neighbors	Query the live roaming domain database from each AP's perspective
Log Analysis	Filter, analyze, and visualize wam.log with 802.11 reason/status code enrichment
Remote Script Executor	Execute shell scripts across an entire AP fleet simultaneously

2 Installation & Getting Started

2.1 Prerequisites

Before deploying Stellar WiFi Analyzer, ensure the following requirements are met:

- **Virtualization platform** (VMware, Hyper-V, Proxmox, or equivalent) — if deploying via the VM option.
- **Docker Engine** installed and running on the host — if deploying via the container option. No Python or system dependencies need to be installed manually; everything is bundled inside the container image.
- **Network connectivity (TCP port 22 / SSH)** from the SWA host to the Stellar WiFi Access Points.
- **A modern web browser** (Chrome, Firefox, Edge, Safari) to access the SWA web interface.
- **Root SSH credentials** for the Stellar APs. Most diagnostic features require root access.

> **Recommended Deployment**
>
> *The VM option is the recommended deployment method. It arrives pre-configured with Docker and the SWA container image already loaded, requiring only a network IP change before use.*

2.2 Install the Application

Option A — Virtual Machine (Recommended)

The VM is the fastest path to a working deployment. It ships as an OVF/VMDK package (VMware / Proxmox) or as a VHD (Hyper-V), with the Alpine Linux OS and the SWA Docker container pre-installed.

- Download the distribution archive:
  - **VMware / Proxmox (OVF + VMDK):** https://drive.google.com/file/d/1IBAhKxW-Bj7izzo5RVACdiEDi_jPC4SO/view
  - **Hyper-V (VHD):** https://drive.google.com/file/d/1ATk4d38xXqAwCjx_2L_YLYi64kB4q1sw/view
- Unzip the archive.
  1. VMWARE/Proxmox: The folder named SWA-VM-OVF will contain the .ovf and one or more .vmdk files.
  2. Hyper-V: The hyper-v disk files will be available
- Import or Create the VM into your virtualization platform:
  1. VMWARE: Import the VM using the standard OVF import process.

2. Hyper-V: Create a new VM using the Generation-1 type and make sure to create 2 disks using the disk files available within the zip file

- VM default resource allocation:
  - 2 vCPUs
  - 8 GB RAM
  - Disk 1: 1 GB (Alpine OS)
  - Disk 2: 32 GB (SWA Application data)
- Power on the VM. The default network configuration is:
  - IP Address: 192.168.13.55 / 24
  - Gateway: 192.168.13.1
  - DNS: 1.1.1.1
- Change the IP address to match your network. Use the hypervisor console to connect to the new VM and edit the file "/etc/network/interfaces" as root (see the credentials few lines below). To edit the file use the command "vi /etc/network/interfaces" and then change the address, netmask and gateway values

```
auto eth0
iface eth0 inet static
    address x.x.x.x
    netmask 255.255.255.0
    gateway x.x.x.1
```

Default SSH credentials for the VM: username root, password G0dL0vesY0u (zeros in place of the letter O).

- Reboot the VM after saving the network configuration. Once rebooted, open your browser and navigate to:

`http://x.x.x.x:3005`

Replace x.x.x.x with the IP address you assigned to the VM.

Option B — Docker Container

If you prefer to run SWA on an existing Linux server or workstation, use the Docker container. The image bundles all dependencies (Python, Flask, Paramiko, ReportLab, and all supporting libraries) — no manual package installation is required.

Pull and start the container with a single command:

```
docker run -d \
  --name log-viewer \
  -p 3005:3005 \
  --restart always \
  fernandorivasplata/log-viewer:latest
```

The -p 3005:3005 flag maps port 3005 on the host to the container. The --restart always flag ensures the container restarts automatically after a host reboot or Docker daemon restart.

2.3 Update the Application

VM Option — Automatic Updates

When running the Stellar WIFI Analyzer (SWA) tool in a VM, updates occur each night at midnight to check for updates. The VM must have internet access and must be operative at that time.

Docker Option — Manual Update

To update to the latest version, run the following three commands in sequence:

```
# Step 1: Pull the latest image
docker pull fernandorivasplata/log-viewer:latest
# Step 2: Stop and remove the running container
docker stop log-viewer && docker rm log-viewer
# Step 3: Start a new container from the updated image
docker run -d --name log-viewer -p 3005:3005 --restart always
fernandorivasplata/log-viewer:latest
```

> ⚠ **Note**
> *Updating removes the old container. Any data or logs stored inside the container will be lost. If you need to preserve session data, save it before updating.*

2.4 Starting the Application

VM Option

The container starts automatically when the VM boots. Simply open a browser and navigate to:

`http://x.x.x.x:3005`

Docker Option

Verify the container is running:

```
docker ps | grep log-viewer
```

You should see log-viewer listed with status Up. Once confirmed, navigate to:

`http://<ipaddress>:3005`

If running Docker on your local machine, use http://localhost:3005 or http://127.0.0.1:3005. If running on a remote server, use that server's IP address.

2.5 Connecting to Access Points

The connection panel at the top of every page accepts up to 10 AP IP addresses. All configured APs are queried in parallel for every diagnostic action.

Field	Description	Example
AP IP 1–10	Management IP addresses of the Stellar APs you want to analyze	192.168.11.36
Username	SSH login username. Root credentials are required for most features	root
Password	SSH login password for the AP	(your AP password)

Pro Tip

Always enter root credentials. While log fetching works with the support account credentials, root access is required for Signal Monitor, CCI/ACI, RTT & Jitter, Multicast/Broadcast, Ethernet Capture, Config Files, and all AP diagnostic functions.

3 Feature Guide

3.1 Device Signal Monitor & CWAP RF Analysis

The Device Signal Monitor tracks any client device's RF performance in real time, polling the AP every 3 seconds and plotting live charts for RSSI, RF Noise Floor, SNR, TX Retransmissions, and Channel Utilization across all three bands simultaneously.

The embedded CWAP-level RF Analysis engine evaluates every sample against industry-standard thresholds and produces a condition assessment (Good / Fair / Poor) with detailed root-cause diagnostics and recommended actions — encoded from 20+ years of wireless troubleshooting expertise.

How to Launch

- Find the device to monitor using the Locate a Device (LD menu) to search for a client by MAC, IP, hostname, RSSI, or SNR and then click on the device or use the Clients Connected option to see every device connected and then click on the Signal link to monitor that specific device.

RF Analysis Checks

Metric	Threshold & Meaning
RSSI	Signal strength graded against MCS-rate thresholds. Below −67 dBm triggers a warning; below −80 dBm is critical.
Noise Floor	Deviation from the −95 dBm baseline. Elevated noise compresses SNR margins and degrades throughput.
SNR	Signal-to-Noise Ratio checked against 802.11 link-budget minimums. Below 20 dB is marginal; below 10 dB is critical.
Channel Utilization	Per-band airtime congestion. Above 68% is warning; above 80% is critical — high CU causes queuing delays.
TX Retransmissions	Real-time retransmission rate. A sustained rate above 15–20% indicates frame loss and link instability.

NEW: Follow Roaming

✨ **New in V1.1.5-build-9.10**

Follow Roaming automatically tracks a client device as it roams between APs, keeping the Signal Monitor active and uninterrupted throughout the roam event.

When a Wi-Fi client roams from one AP to another, a standard Signal Monitor session would lose visibility the moment the client disassociates from the original AP. Follow Roaming eliminates this gap by automatically detecting the roam event and seamlessly switching the monitor context to whichever AP the client has associated with next.

This is invaluable for diagnosing roaming-related problems such as:

- Sticky clients that roam too late — you can observe the RSSI deterioration before the roam event and the signal improvement (or lack thereof) after.
- Roam-loop conditions — where the client bounces repeatedly between two APs, visible as a rapid sequence of Follow Roaming transitions.
- Roaming failures — where the expected AP transition does not complete, leaving the client in a degraded RF state.
- Post-roam RF quality validation — confirming the client landed on a better AP with improved RSSI and SNR.

Follow Roaming operates by continuously monitoring the association table across all configured APs. When the target device MAC is detected on a different AP than the one currently being polled, the monitor transparently switches polling to the new AP and annotates the chart timeline with a roam event marker.

> ℹ️ **Tip**
>
> *For the most complete roaming analysis, enter all APs in the site (or at least all neighboring APs) in the connection panel before launching the monitor. Follow Roaming can only track roams to APs that are in your configured AP list.*

NEW: Anomaly Detection — Summary Panel & Filters

> ✨ **New in V1.1.5-build-9.10**
>
> *The Signal Monitor now includes a real-time Anomaly Summary panel that tracks, counts, and categorizes RF anomalies detected during the monitoring session, with filter controls to focus on specific anomaly types.*

As the Signal Monitor runs, the RF Analysis engine evaluates each 3-second sample and classifies any out-of-threshold condition as an anomaly. These anomalies are now aggregated in the Anomaly Summary panel, which provides:

- **Event count per anomaly type** — see at a glance how many times RSSI dropped below threshold, noise spiked, SNR degraded, or channel utilization exceeded limits.
- **Severity coloring** — each anomaly type is color-coded (green / yellow / red) matching the severity of the worst observed condition for that metric.
- **Timeline filter** — click any anomaly type to filter the chart view and highlight only the periods where that specific condition occurred, making it easy to correlate RF degradation with user complaint timestamps.
- **Session anomaly score** — an overall health score for the monitoring session, calculated from the frequency and severity of all detected anomalies.

Anomaly types tracked:

Anomaly	Trigger Condition
Weak Signal	RSSI drops below −67 dBm (warning) or −80 dBm (critical)
Elevated Noise	Noise floor rises above −88 dBm
Low SNR	SNR falls below 20 dB (warning) or 10 dB (critical)
High Channel Utilization	CU exceeds 68% (warning) or 80% (critical) on any band
High Retransmissions	TX retransmission rate sustained above 15%
Roam Event	Client detected on a different AP (Follow Roaming trigger)

> **Tip**
> *Save the Signal Monitor session after a monitoring period and replay it later via RA → Monitor Replay → Device Signal Monitor Replay. All anomaly data is preserved in the saved JSON file.*

3.2 Channel Utilization Monitor

The Channel Utilization Monitor shows exactly how congested each radio band is in real time. High channel utilization is one of the most common — and least visible — causes of Wi-Fi complaints including slow speeds, high latency, and intermittent disconnections.

How to Use

- Enter AP IP(s) and root credentials in the connection panel.
- Go to RA → Channel Utilization.
- The fullscreen view opens with gauge dials and time-series charts for all three bands.
- Click Start Monitoring to begin continuous polling every 3 seconds.
- Click Save Session to export data as a JSON file for replay.

Utilization	Status	Action
0–68%	Healthy (green)	Normal operation. Airtime is available for all clients.
69–79%	Warning (yellow)	Channel becoming congested. Consider band steering or load balancing.
80–100%	Critical (red)	Near-saturated. Immediate action required — add APs or adjust design.

3.3 CCI & ACI Interference Detection

Performs a full 802.11 neighbor scan to discover every nearby AP causing Co-Channel Interference (CCI) and Adjacent-Channel Interference (ACI). Results are signal-weighted so you see real severity — not just a count.

As of V1.1.5-build-9.10, the CCI scan also measures the ambient noise floor on each 2.4 GHz channel (1, 6, and 11) and factors it into the channel recommendation, giving you a more complete picture of which channel is truly the cleanest option.

How to Use

- Enter AP IP(s) and root credentials.
- Go to RA → CCI & ACI and select the target band when prompted.
- Review the CCI gauge, ACI count, signal-weighted impact %, and channel recommendation.
- Optionally click Start CCI-ACI Monitor to track counts continuously over time.

Result Item	Description
CCI Gauge	Number of co-channel APs competing for your airtime on the same channel.
ACI Count	Adjacent-channel APs whose signal bleeds into your operating channel.
Signal-Weighted Impact %	RSSI-weighted load score. A single strong nearby AP causes more impact than ten distant ones.
2.4 GHz Channel Recommendation	Ranks Ch 1/6/11 by CCI count + noise floor and highlights the least-congested option.
Per-Channel Noise Floor	Measured ambient noise in dBm for each 2.4 GHz channel. Color-coded: green (≤−95 dBm), yellow (−94 to −89 dBm), red (> −88 dBm).
Detected Radio Table	BSSID, SSID, channel, signal strength, CCI/ACI impact, and last-seen timestamp.

3.4 Locate a Device

Searches for any connected client across all configured APs simultaneously — even when you don’t know which AP the device is on. The fastest path to the Signal Monitor.

Five Search Modes

Mode	Description	Example
By MAC Address	Full or partial MAC address match	a4:c3 → matches all devices with that OUI prefix

Mode	Description	Example
By IP Address	Full or partial IP address match	192.168.1 → lists all devices on that subnet
By Hostname	Device name or partial match	iphone → matches all iPhones
By RSSI ≤	All clients below a signal threshold	75 → finds devices at −75 dBm or worse
By SNR ≤	All clients below an SNR threshold	15 → devices suffering from a noisy link

How to Use

- Go to LD → Locate a Device.
- Select a search mode tab, enter your search value, click Search.
- Results show MAC, IP, hostname, associated AP, band, RSSI, and SNR.
- Click any result row to instantly launch the Signal Monitor for that device.

3.5 RTT & Jitter Monitor

✨ **New in V1.1.5-build-9.10**

The RTT & Jitter Monitor is a new Traffic Analysis function that measures round-trip latency and jitter from one or more APs to any target host, providing a live view of network path quality.

Network latency and jitter are invisible to most Wi-Fi diagnostic tools, yet they are often the root cause of complaints about voice call quality, video conferencing instability, and sluggish application response times. The RTT & Jitter Monitor closes this gap by running continuous ICMP ping measurements directly from the AP to any destination you specify — a gateway, a DNS server, a cloud endpoint, or any IP reachable from the AP.

Why It Matters

- **Baseline vs. problem state comparison** — measure RTT during normal operation and during a reported issue to quantify the degradation.
- **Path quality from the AP perspective** — because the measurement originates from the AP itself, it captures problems in the wired uplink and core network, not just the wireless segment.
- **Jitter analysis for real-time traffic** — VoIP and video conferencing are highly sensitive to jitter. RTT variation above 20–30 ms typically causes perceptible audio/video quality degradation.
- **Multi-AP simultaneous measurement** — run the same test from multiple APs at once to identify whether a latency problem is localized to a specific AP’s uplink or is systemic.

How to Use

- Go to TA → RTT & Jitter Monitor.
- Select the target AP (or multiple APs) from the AP selector.
- Enter the destination host IP or hostname (e.g., 8.8.8.8, your gateway IP, or a server name).
- Set the polling interval (default: 1 second).
- Click Start Monitoring. Live RTT and jitter values are plotted in real time.
- Click Save Session to export the data as JSON for later analysis.

Metric	Description
RTT (Round-Trip Time)	Time in milliseconds for a ping packet to travel from the AP to the target and back. Plotted as a live time-series.
Jitter	Variation between consecutive RTT measurements (|RTT_n − RTT_{n-1}|). High jitter indicates an unstable path.
Packet Loss	Percentage of ping packets that received no response. Any sustained loss indicates a serious path issue.
Min / Avg / Max RTT	Statistical summary of the measurement session, updated continuously.

RTT Threshold	Interpretation
< 10 ms	Excellent — local network path, very low latency
10–50 ms	Good — typical for LAN-to-internet paths
50–100 ms	Acceptable — may impact real-time applications
> 100 ms	Poor — investigate uplink congestion, routing, or WAN issues

Tip

Pair RTT & Jitter Monitor with the Channel Utilization Monitor. If RTT is high only when CU is also high, the bottleneck is RF congestion on the AP's radio — not the wired uplink. If RTT is high even at low CU, the issue is upstream of the AP.

3.6 Multicast / Broadcast Rate Monitor

✨ **New in V1.1.5-build-9.10**

The Multicast/Broadcast Rate Monitor is a new Traffic Analysis function that measures the rate of multicast and broadcast frames on the AP's wireless interfaces, with separate thresholds for each traffic type and an intelligent alert system.

Excessive multicast and broadcast traffic is one of the most common but hardest-to-diagnose causes of Wi-Fi degradation. Unlike unicast traffic, multicast and broadcast frames are transmitted at the lowest mandatory data rate (often 1 or 6 Mbps) and cannot be acknowledged by clients — meaning they consume a disproportionate amount of airtime relative to their payload. On busy networks, uncontrolled multicast/broadcast traffic can saturate the channel and crowd out legitimate client traffic.

Measured Traffic Types

Traffic Type	Source	Typical Culprits
Multicast RX (pps)	Frames received by the AP's wireless interface from multicast sources	mDNS, Bonjour, SSDP, IGMP, routing protocols
Broadcast RX (pps)	Broadcast frames received by the AP from the wired or wireless side	ARP storms, DHCP broadcasts, NetBIOS announcements
Broadcast TX (pps)	Broadcast frames transmitted by the AP to its associated clients	ARP requests flooded to all clients, spanning tree BPDUs

Note: Linux kernel network statistics do not expose a per-interface Multicast TX counter (there is no tx_multicast equivalent in sysfs). The Multicast TX metric is therefore not available. All other metrics are measured directly from the AP's kernel interface statistics.

Alert Thresholds

The monitor applies separate thresholds for each traffic type, reflecting the different impact each has on the wireless medium:

Traffic Type	Warning Threshold	Critical Threshold	Impact When Exceeded
Multicast RX	50 pps	200 pps	Airtime consumed by low-rate multicast delivery to all clients; can crowd out unicast traffic
Broadcast RX	100 pps	500 pps	ARP storms or DHCP floods; degrades performance for all associated clients
Broadcast TX	200 pps	700 pps	AP is flooding broadcasts to all clients; highest airtime impact per frame on 802.11

Alert Panel & Remediation Guidance

When any threshold is crossed, the monitor displays an Alert Panel with:

- **Impact description** — a plain-language explanation of what the elevated traffic rate means for your wireless clients.
- **Remediation steps** — specific actions to reduce the traffic: enabling IGMP snooping, enabling ARP proxy on the AP, implementing broadcast suppression policies, identifying the source hosts causing the floods, and reviewing VLAN design to limit broadcast domain size.
- **Severity level** — Warning (yellow) or Critical (red), based on which threshold was crossed.

Alerts can be dismissed individually. A new alert fires only when the severity level increases above the previously dismissed level, preventing alert fatigue during a monitoring session.

How to Use

- Go to TA → Multicast/Broadcast Monitor.
- Select the target AP from the AP selector.
- Click Start Monitoring. Multicast RX, Broadcast RX, and Broadcast TX rates are plotted in real time (pps).
- Review the alert panel if any threshold is crossed. Follow the remediation steps displayed.
- Use the color-coded legend to interpret rate levels: green (normal), yellow (warning), red (critical).

3.7 Ethernet Capture (tcpdump)

Runs a live tcpdump packet capture directly on AP ethernet interfaces. Packets stream into the browser in real time with no physical access required.

How to Use

- Go to TA → Ethernet Capture.
- Check the AP(s) to capture from and select the interface (eth0, eth1, or custom).
- Choose a pre-built filter (DHCP, DNS, ARP, HTTP) or enter a custom BPF expression.
- Set the auto-stop timer (1–1440 minutes). Default is 5 minutes.
- Click Start Capture. Live packets stream into the output area.
- Click Save All PCAPs to download .pcap files for Wireshark analysis.

Note

Auto-stop can be set from 1 minute up to 1440 minutes (24 hours) for long-duration captures. Always respect applicable network monitoring policies.

3.8 AP Information Dashboard

Delivers a complete hardware and configuration snapshot of every configured AP. Up to 10 APs can be displayed side by side. Requires root credentials.

Information Panel	Contents
Identity	AP Name, Model, Firmware version, Country Code, Mode, OV Controller address
Network	Management IP, Ethernet Speed, Duplex mode (Full / Mesh)
Radio Configuration	Active channels and MIMO configuration for 2.4 / 5 / 6 GHz radios
SSIDs & Radios	Per-radio SSID assignments and authentication type
Hardware Health	Free CPU %, free memory, free disk, CPU temperature
VLANs & Profiles	Required trunk VLANs and profile assignments

3.9 Config Files Browser

✨ **New in V1.1.5-build-9.10**

The Config Files Browser lets you navigate, view, and download configuration and system files directly from any Stellar AP's file system, without needing a separate SSH session.

When troubleshooting an AP, the ability to quickly inspect its configuration files — without opening a separate terminal and navigating the shell manually — saves significant time. The Config Files Browser provides a clean, browser-based interface to browse the AP's file system, view file contents with syntax highlighting, and download files to your local machine for offline analysis or archiving.

Key Capabilities

- **Directory navigation** — browse the AP's file system tree from the browser. Navigate to any directory the AP's SSH user can access.
- **File content viewer** — view configuration files, logs, and system files inline, with line numbers and readable formatting.
- **File download** — download any file to your local machine with a single click. Useful for archiving configuration snapshots before/after changes, or sending files to a colleague for review.

- **Multi-AP support** — switch between any of the configured APs using the AP selector, allowing you to compare configuration files across multiple APs in quick succession.

How to Use

- Go to RA → Config Files Browser.
- Select the target AP from the AP selector.
- The file browser opens showing the root directory. Click any folder to navigate into it.
- Click any file to view its contents in the inline viewer.
- Click the Copy button to save the file to your local machine.

> ⚠ **Note**
> *The Config Files Browser requires root credentials to access most AP system directories. Use it responsibly and in accordance with your organization's change management and security policies.*

3.10 Roaming Neighbors

The Roaming Neighbors function queries the AP's internal roaming domain database over SSH and displays a structured table of every AP that the queried AP is aware of as a potential roaming neighbor. This reflects the live state of the roaming topology as seen from each AP's perspective.

Understanding the roaming neighbor table is essential for diagnosing roaming failures, sticky-client problems, and unexpected AP transitions. It answers the critical question: Does this AP know about its neighbors, and at what signal level does it see them?

Why It Matters

- **Roaming topology validation** — confirms which APs are visible to each other and whether expected neighbor relationships exist. Missing neighbors are a common root cause of poor roaming.
- **Signal-based roaming intelligence** — each entry shows the RSSI at which the AP hears the neighbor. Color-coded thresholds immediately show which neighbors are strong handoff candidates vs. marginal connections.
- **Multi-AP comparative view** — with multiple APs configured, see the neighbor table from every AP simultaneously — allowing side-by-side comparison of topology consistency.
- **Pre-deployment & post-change verification** — after channel plan changes, AP replacements, or firmware updates, verify all expected neighbors are present at acceptable RSSI.

Column	Description
AP Name / BSSID	Identifier of the neighboring AP

Column	Description
RSSI	Signal level at which this AP hears the neighbor (dBm). Green ≥ −67 dBm, yellow −68 to −80 dBm, red < −80 dBm.
Channel	Radio channel the neighbor is operating on
State	Neighbor relationship state in the roaming domain
Last Seen	How recently the neighbor was last detected

3.11 Remote Script Executor

The Remote Script Executor is accessible via the RS menu. It lets you write or load any shell script and execute it across an entire fleet of Stellar APs simultaneously — turning individual manual SSH sessions into a single coordinated operation that completes in seconds.

Why It Is Important

- **Fleet-wide operations in seconds** — execute any shell command or multi-line script across all your APs in parallel.
- **Consistent diagnostics** — collect the same information from every AP simultaneously — logs, configuration values, interface states.
- **Script library** — save frequently used scripts by name for instant reuse.
- **Safe & clean execution** — scripts are uploaded, executed, and automatically deleted after completion. Unreachable APs are skipped gracefully.

Supported Target IP Formats

Format	Example	Description
Single IP	192.168.11.36	One specific AP
Comma-separated	192.168.11.36, 192.168.11.43	Multiple specific APs
IP Range	192.168.11.36-50	All IPs from .36 to .50
CIDR Block	192.168.11.0/24	All hosts in the subnet
Mixed	192.168.1.1, 10.0.0.0/24	Combine any formats freely

Script Upload Method Fallback Chain

- SFTP — Direct file transfer via SSH SFTP subsystem. Fastest, preferred method.
- SCP — Secure copy via SCP protocol if SFTP is unavailable.
- Base64 exec — Encodes the script as base64 and pipes it via shell exec. Works even on APs where file transfer protocols are restricted.

⚠ **Important**

The Remote Script Executor requires root credentials. All scripts run as root on the AP. Always test a new script on a single AP before running it fleet-wide.

4 Advanced Functions

4.1 Log Analysis & Network Events Analyzer

Reads /var/log/wam.log from each AP with powerful filtering and automatic 802.11 reason/status code enrichment. The Network Events Analyzer reconstructs per-client connection flows and detects roam loops, RADIUS failures, VLAN anomalies, and DHCP timeouts.

Click any MAC address in log output to instantly filter the log and open the Network Events Analyzer for that specific device.

<table>
<tr><th>Filter Expression</th><th>Meaning</th></tr>
<tr><td>keyword</td><td>Lines containing this word (case-insensitive)</td></tr>
<tr><td>A & B</td><td>Lines containing both A AND B</td></tr>
<tr><td>A | B</td><td>Lines containing A OR B</td></tr>
<tr><td>!keyword</td><td>Exclude lines containing this word</td></tr>
<tr><td>(dhcp|roaming)&!probe</td><td>Combine operators freely for complex filters</td></tr>
</table>

4.2 Monitor Mode

Starts a time-aligned continuous log capture across all APs simultaneously. New lines are appended every 60 seconds. Maximum session: 30 minutes. On stop, all files are packaged into a ZIP archive for download.

Monitor Mode is ideal for capturing intermittent events: run it during a reported issue window, stop it when the event ends, and analyze the downloaded logs with the Log Analysis tool.

5 Tips, Best Practices & Troubleshooting

01

Always use root credentials

Most diagnostic features require root SSH access. Log fetching works with the support credentials, but root unlocks the full toolset including Signal Monitor, CCI/ACI, RTT & Jitter, Multicast/Broadcast, and Config Files.

02

Enter multiple APs for comparative analysis

Enter up to 10 AP IPs to query all of them in parallel — essential for diagnosing roaming issues, comparing RF conditions across a site, or running fleet-wide scripts. Start with 2 or 3 APs and then grow if needed.

03

Use Follow Roaming for sticky-client diagnosis

Open the Device Signal Monitor before asking a client to walk through the coverage area. Watch for late roam events (RSSI already below −75 dBm at roam time) and post-roam RF quality to diagnose sticky-client behavior.

04

Use the MAC link in log output

Every MAC address in log output is a clickable link. Clicking it filters the log and opens the Network Log Events Analyzer for that device — the fastest path from a log event to a per-client timeline.

05

Save Signal Monitor sessions

For intermittent issues, run the Signal Monitor for 30+ minutes, save the JSON, and replay later via RA → Monitor Replay. All RF data and anomalies are preserved.

06

Use date/time filter for incident analysis

Enable "Show logs from" to focus exclusively on a specific incident window, avoiding noise from unrelated activity before or after the event.

07

Ethernet Capture for Layer 2/3 issues

When a client associates but cannot get an IP, use Ethernet Capture with the DHCP filter on eth0 to pinpoint whether the DHCP Discover is reaching the switch, whether the server is responding, and whether the Offer is reaching the AP.

08

RTT & Jitter before blaming Wi-Fi

When users complain about application latency, run the RTT & Jitter Monitor from the AP to the default gateway. If RTT is high even at low Channel Utilization, the problem is in the wired network — not the wireless.

09 **Multicast/Broadcast Monitor during broadcast storms**

If clients report intermittent connectivity, run the Multicast/Broadcast Monitor. A broadcast RX rate above 500 pps (critical) is a strong indicator of an ARP storm or DHCP loop impacting wireless performance.

10 **CCI scan before channel plan changes**

Always run a CCI & ACI scan before changing channels. The channel recommendation now includes per-channel noise floor measurement, giving you the most complete picture of which 2.4 GHz channel is truly the cleanest option.

11 **Check AP health regularly**

Use AP Information to monitor CPU temperature and memory. APs above 70°C or below 10% free CPU may exhibit stability or performance issues.

12 **Config Files Browser for pre/post change archiving**

Before making any configuration change on an AP, use the Config Files Browser to download the relevant configuration files. This gives you a baseline to compare against if issues arise after the change.

Stellar WiFi Analyzer — V1.1.5-build-9.10

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CWNA | CWAP | CWSP — 20+ Years of Wi-Fi Expertise

Soli Deo Gloria